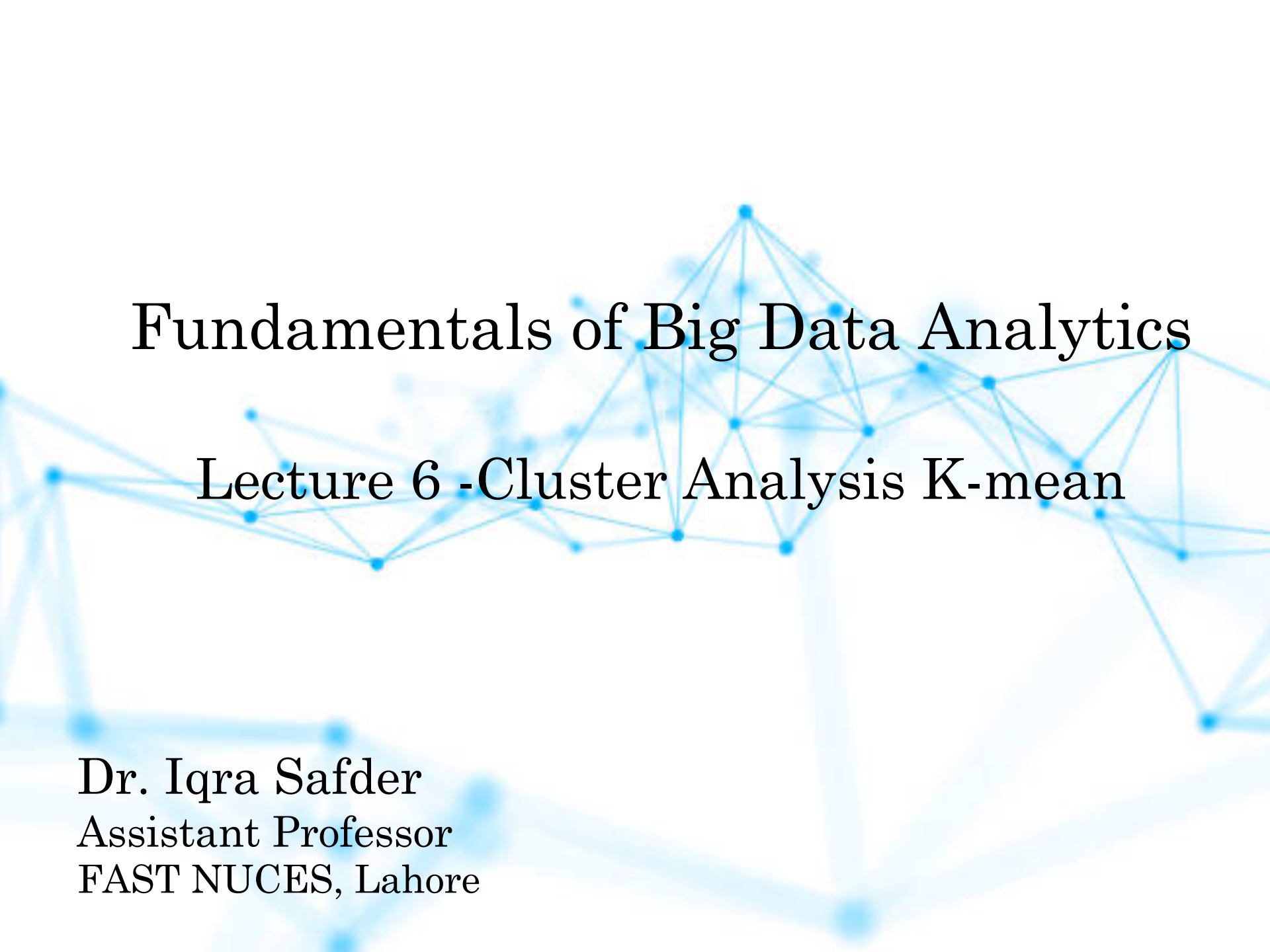




A faint, light blue background graphic consisting of a network of interconnected nodes and lines, resembling a data graph or a molecular structure, spanning the entire slide.

Fundamentals of Big Data Analytics

Lecture 6 -Cluster Analysis K-mean

Dr. Iqra Safder
Assistant Professor
FAST NUCES, Lahore

Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering



K-means Clustering

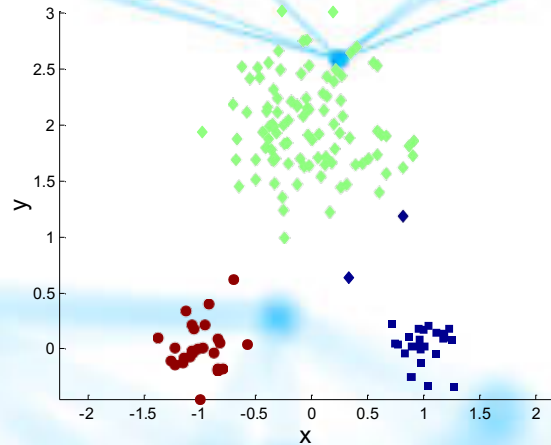
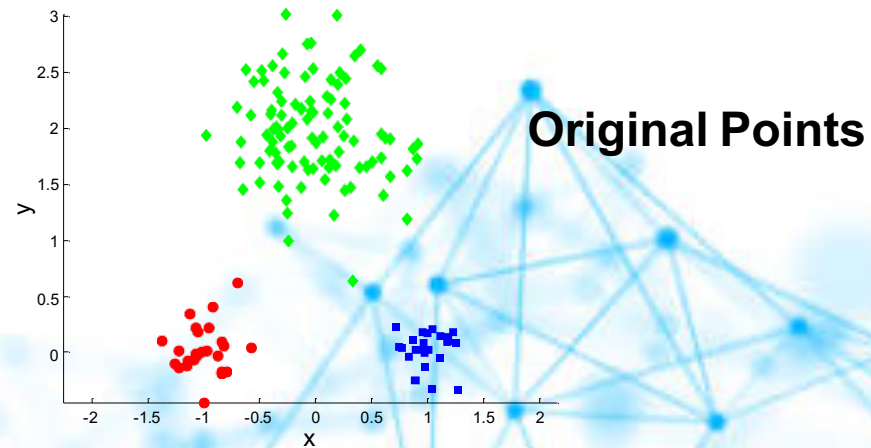
- Partitional clustering approach
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- Number of clusters, K , must be specified
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

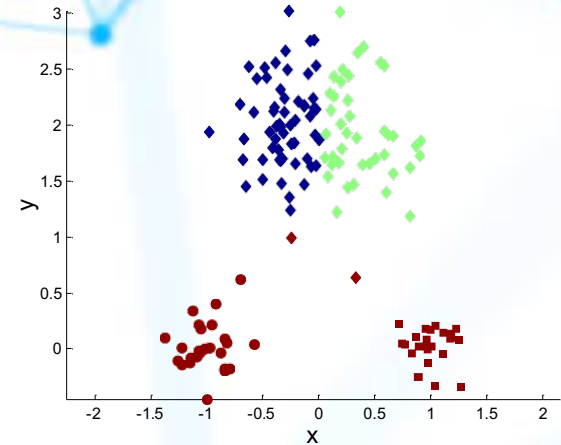
K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- 'Closeness' is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

Two different K-means Clusterings

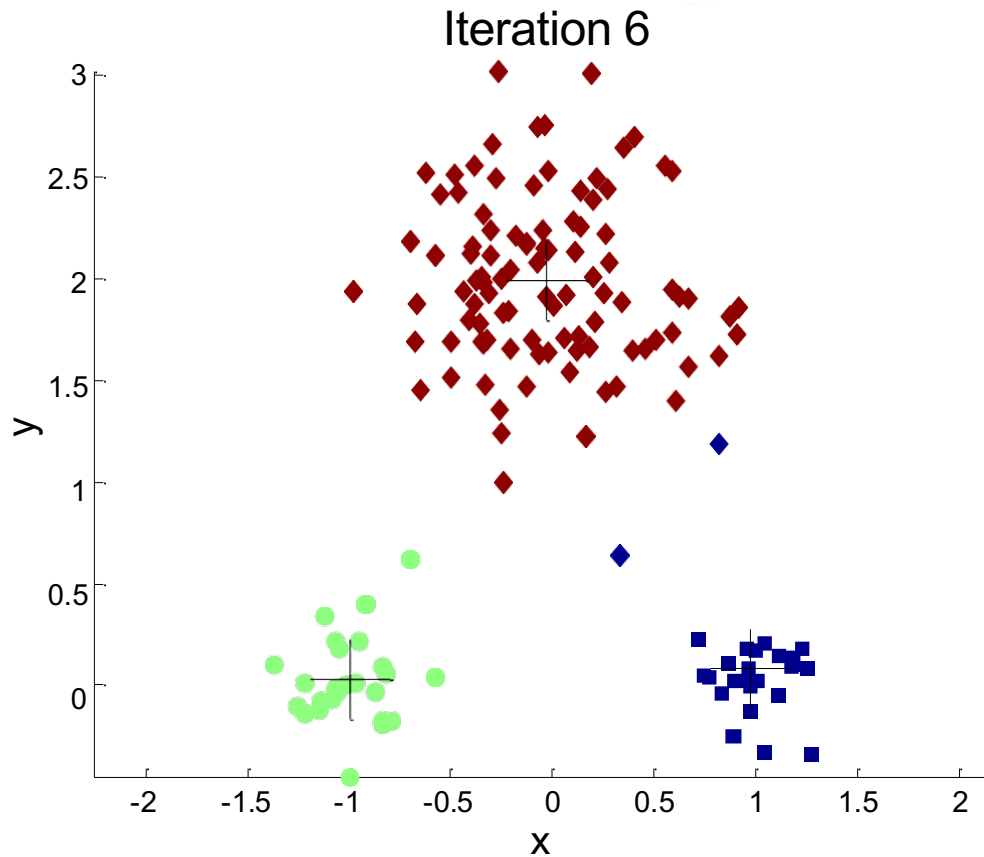


Optimal Clustering

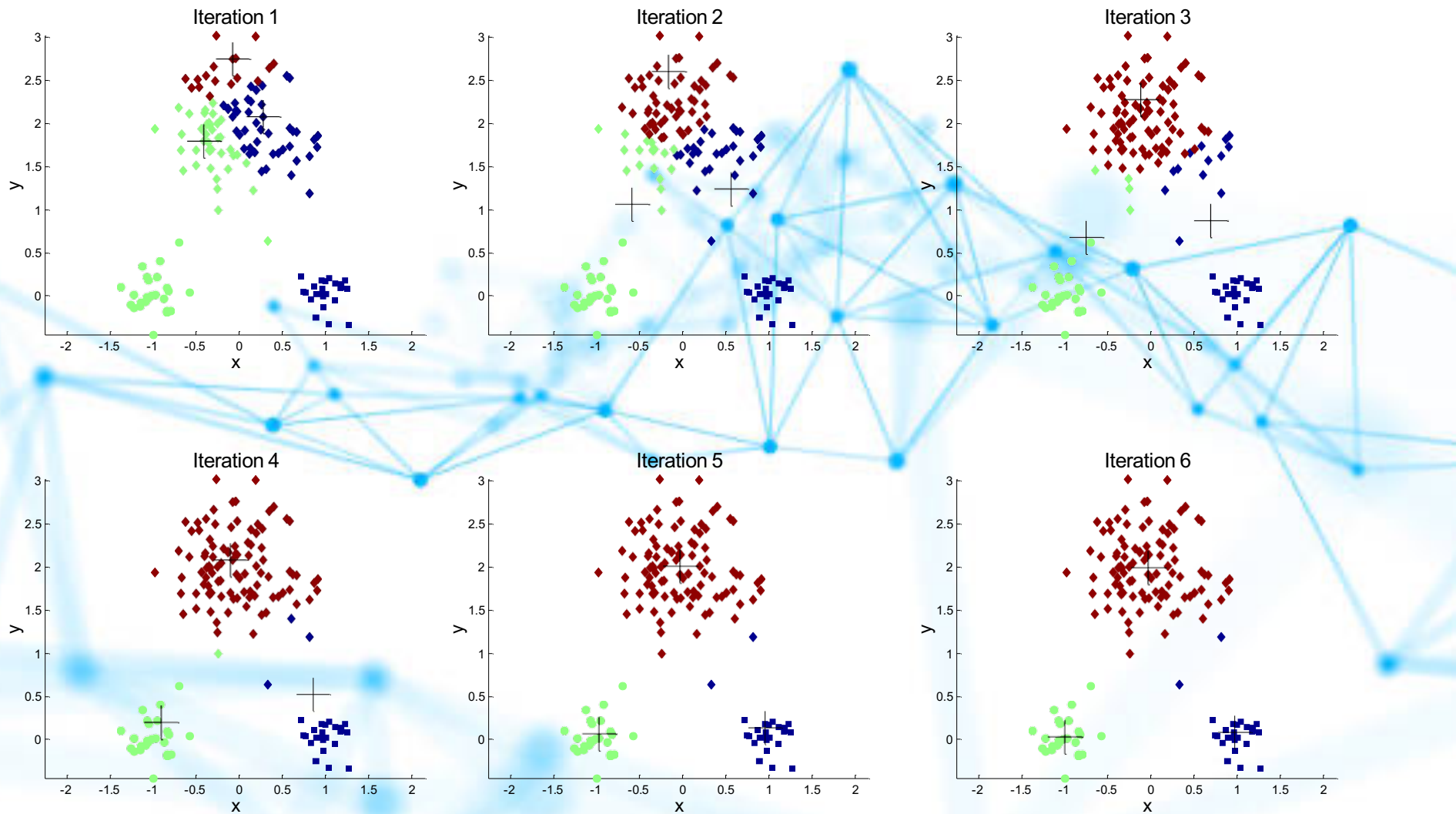


Sub-optimal Clustering

Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



Partitioning Algorithms

Partitioning method: Construct a partition of n documents into a set of K clusters

Given: a set of documents and the number K

Find: a partition of K clusters that optimizes the chosen partitioning criterion

- Globally optimal

- Intractable for many objective functions

- Effective heuristic methods: K -means algorithm

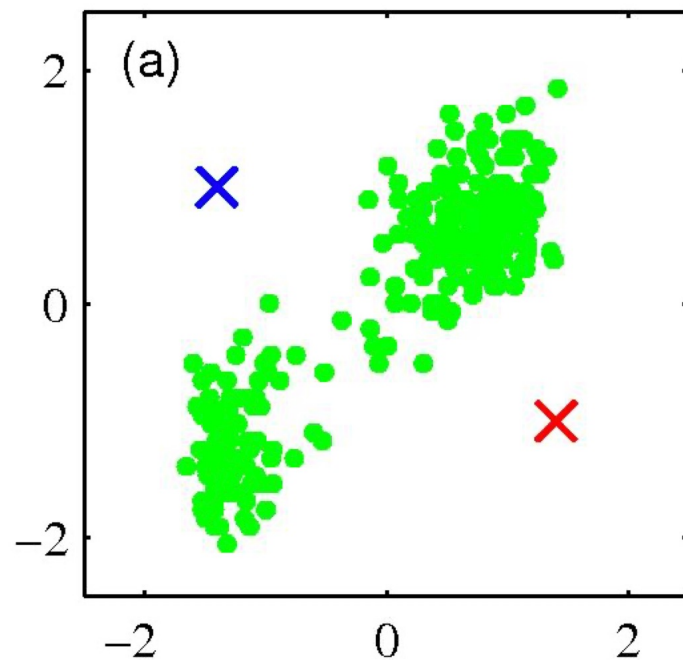
K-Means

- Assumes documents are real-valued vectors e.g. with TF-IDF.
- Clusters based on *centroids* (aka the *center of gravity* or mean) of points in a cluster, c :

$$\vec{\mu}(c) = \frac{1}{|c|} \sum_{\vec{x} \in c} \vec{x}$$

- Reassignment of instances to clusters is based on distance to the current cluster centroids.

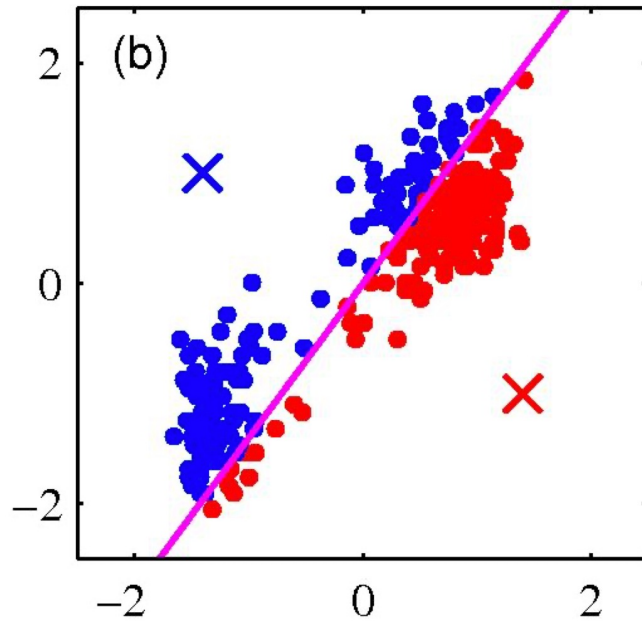
K-Means



Pick K random points as cluster centres (means)

Shown here for $K=2$

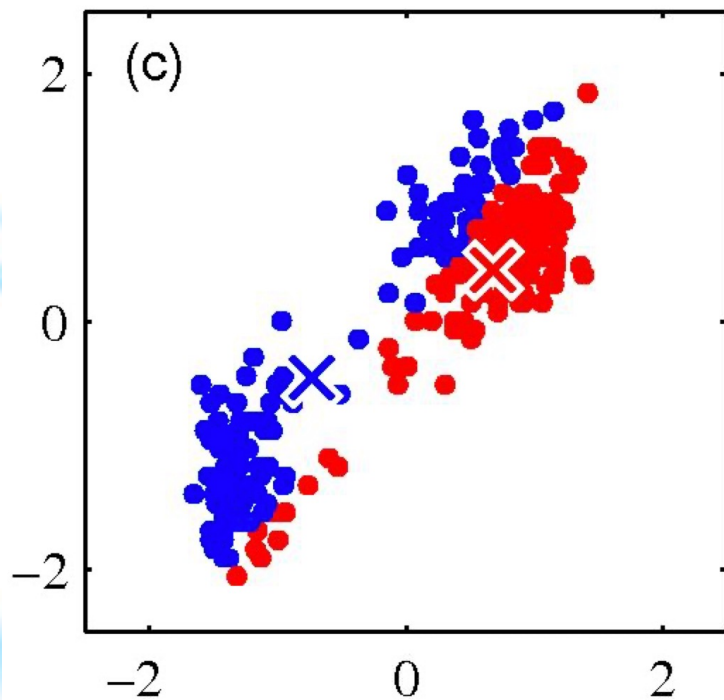
K-Means



Iterative Step 1

- Assign data points to closest cluster centre

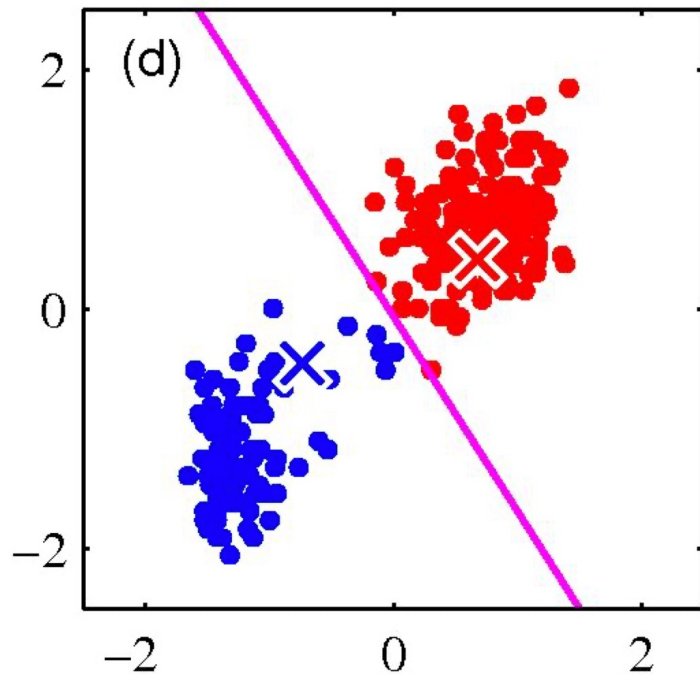
K-Means



Iterative Step 2

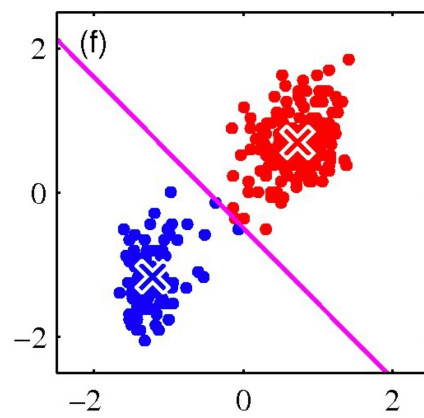
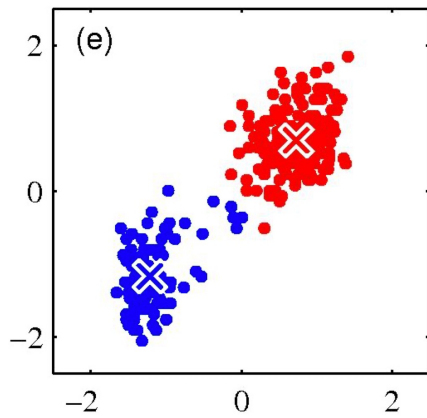
- Change the cluster centre to the average of the assigned points

K-Means

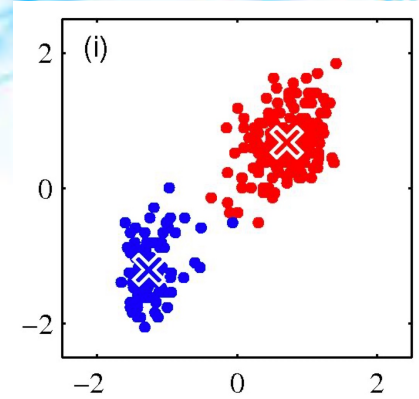


Repeat until convergence

K-Means

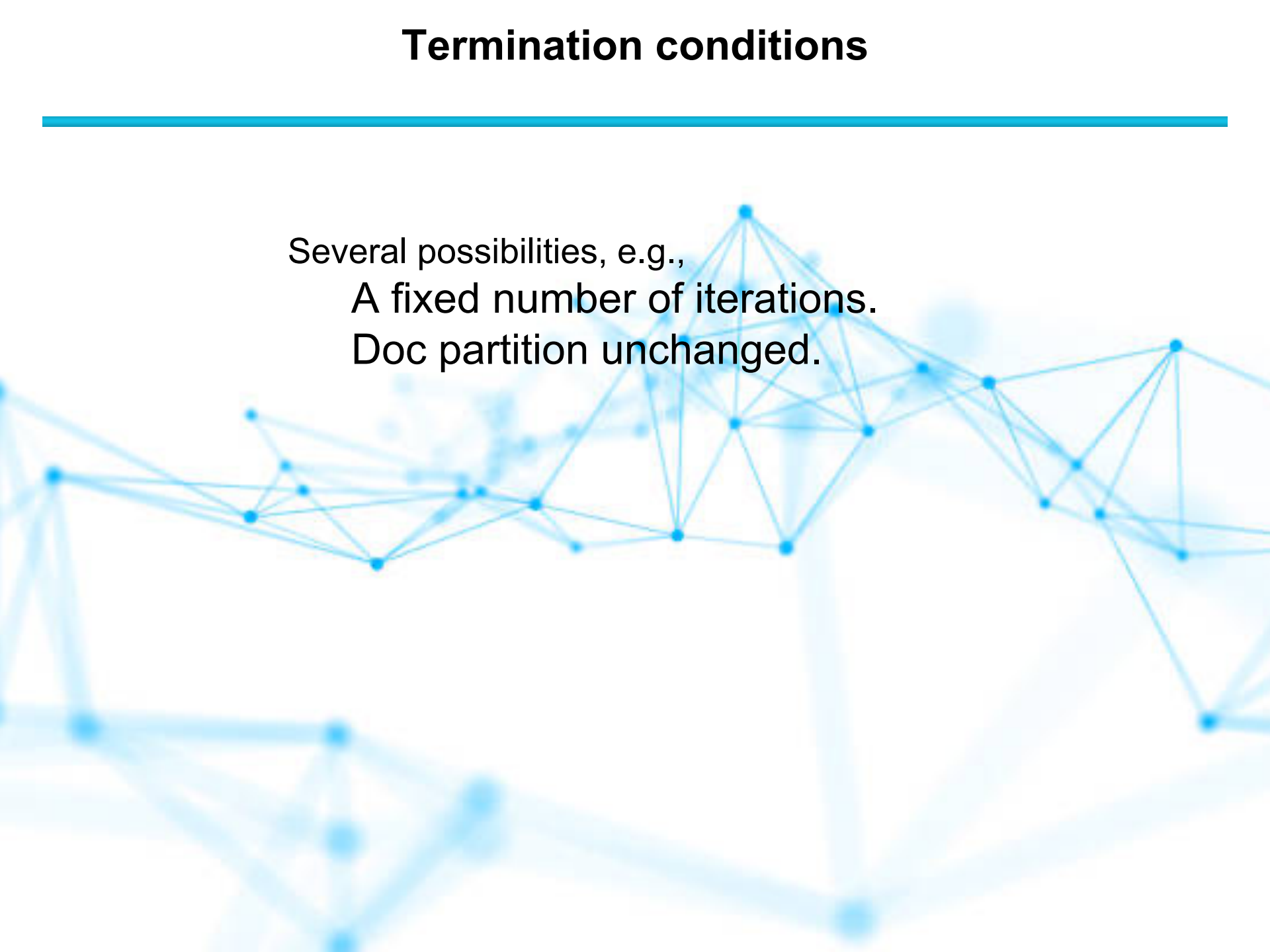


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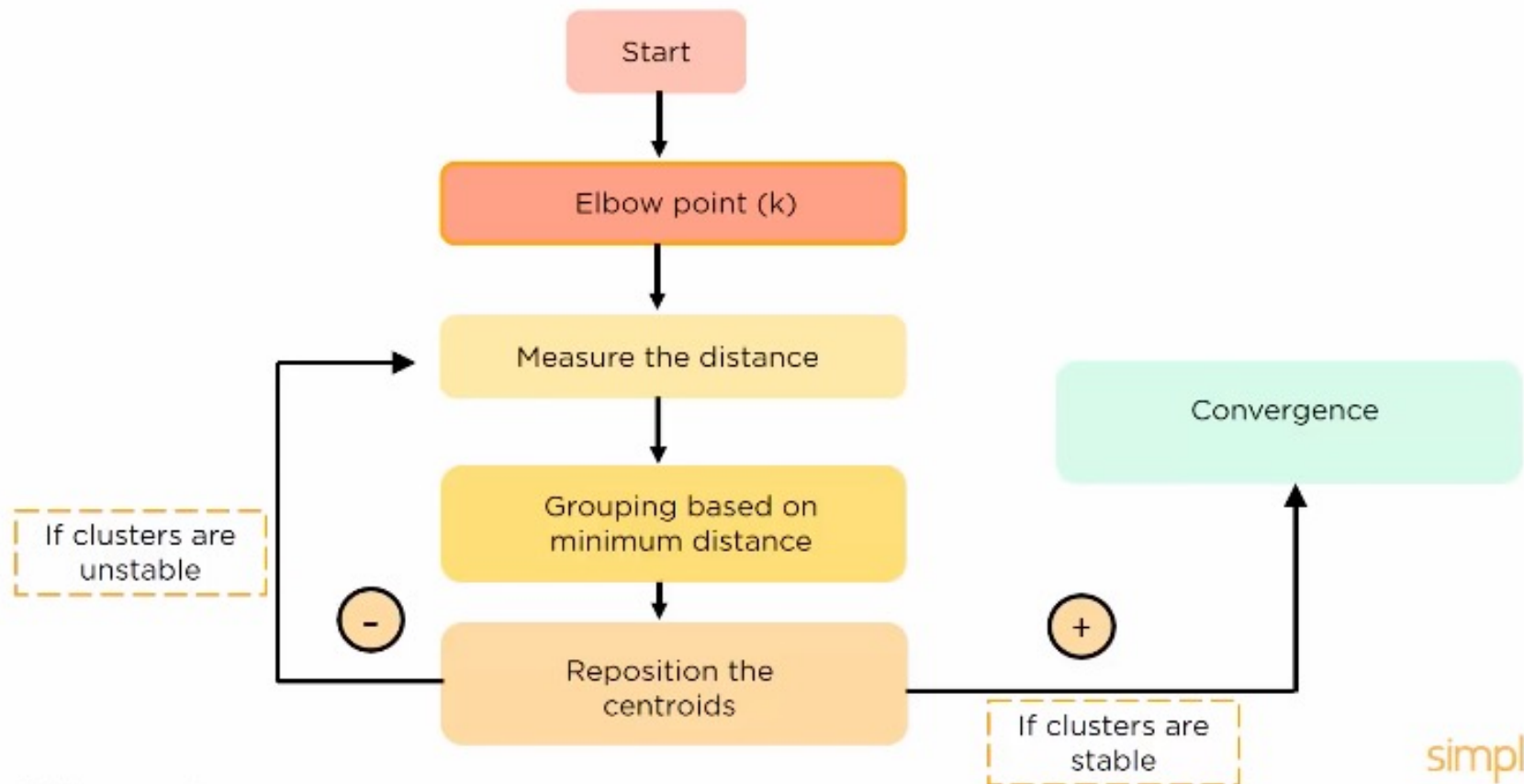


Termination conditions

Several possibilities, e.g.,
A fixed number of iterations.
Doc partition unchanged.



How does K-Means clustering work?



How does K-Means clustering work?

Elbow point

Measure the distance

Grouping

Reposition the centroids

Convergence

- Let's say, you have a dataset for a Grocery shop



- Now, the important question is, *"how would you choose the optimum number of clusters?"*



How does K-Means clustering work?

Elbow point

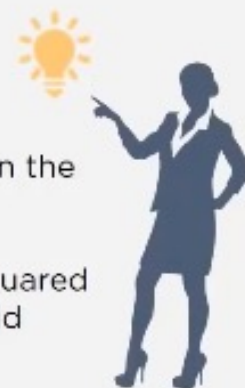
Measure the distance

Grouping

Reposition the centroids

Convergence

- The best way to do this is by **Elbow method**
- The idea of the elbow method is to run K-Means clustering on the dataset where 'k' is referred as number of clusters
- Within sum of squares (WSS) is defined as the sum of the squared distance between each member of the cluster and its centroid



$$WSS = \sum_{i=1}^m (x_i - c_i)^2$$

Where x_i = data point and c_i = closest point to centroid

How does K-Means clustering work?

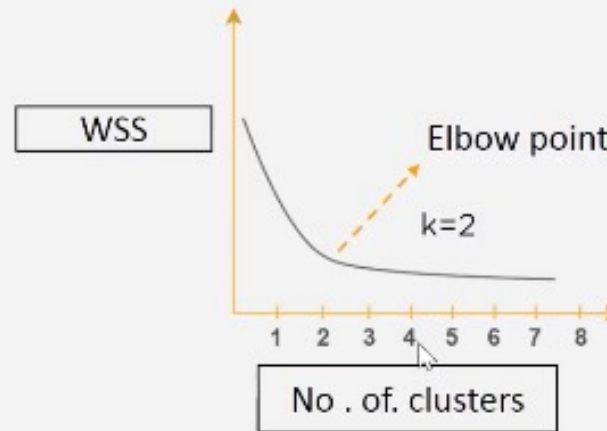
Elbow point

Measure the distance

Grouping

Reposition the centroids

Convergence



- Now, we draw a curve between *WSS* (within sum of squares) and the *number of clusters*
- Here, we can see a very slow change in the value of WSS after $k=2$, so you should take that elbow point value as the final number of clusters

Evaluating K-means Clusters

- Most common measure is **Sum of Squared Error (SSE)**

- For each point, the error is the distance to the nearest cluster
- To **get SSE**, we **square these errors and sum them.**

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the representative point for cluster C_i
 - ◆ can show that m_i corresponds to the center (mean) of the cluster
- Given two clusters, we can choose the one with the smallest error
- One easy way to reduce SSE is to **increase K**, the number of clusters
 - ◆ **A good clustering with smaller K can have a lower SSE than a poor clustering with higher K**

How does K-Means clustering work?

Elbow point

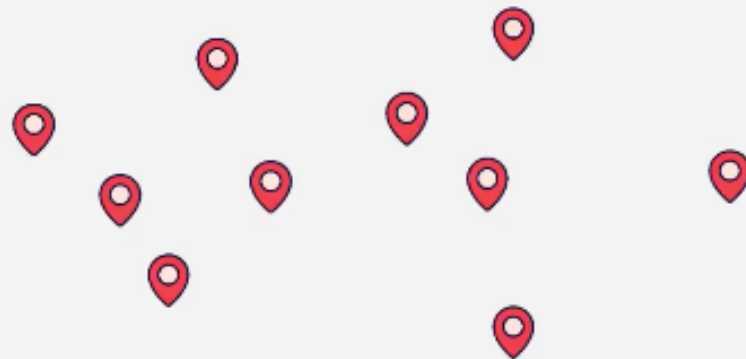
Measure the distance

Grouping

Reposition the centroids

Convergence

Step 1: The given data points below are assumed as **delivery points**



How does K-Means clustering work?

Elbow point

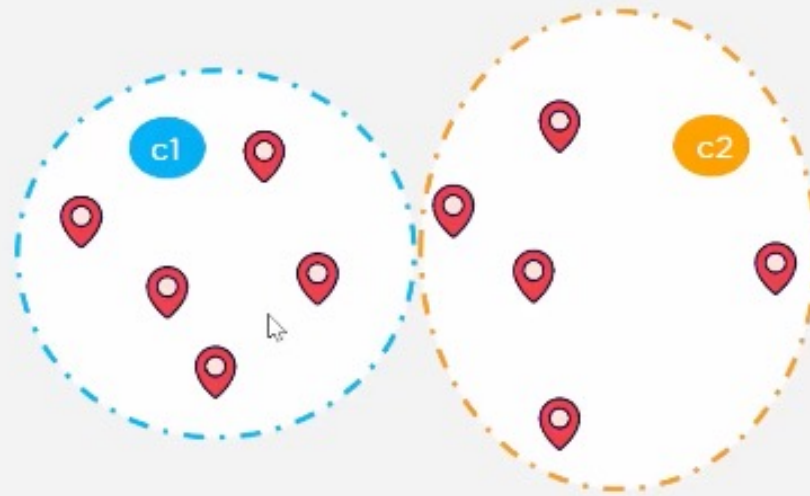
Measure the distance

Grouping

Reposition the centroids

Convergence

Step 3: Based upon the distance from c1 and c2 centroids, the data points will group itself into clusters



How does K-Means clustering work?

Elbow point

Measure the distance

Grouping

Reposition the centroids

Convergence

Step 6: Now, compute the centroid of data points inside the orange cluster

Step 7: Reposition the centroid of the orange cluster to the new centroid



How does K-Means clustering work?

Elbow point

Measure the distance

Grouping

Reposition the centroids

Convergence

Step 6: Now, compute the centroid of data points inside the orange cluster

Step 7: Reposition the centroid of the orange cluster to the new centroid



A Simple Example of k-means working

□ Let $K=2$

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

A Simple Example of k-means working

Step 1:

Initialization: Randomly we choose following two centroids ($k=2$) for two clusters. In this case the 2 centroid are: $m_1=(1.0,1.0)$ and $m_2=(5.0,7.0)$.

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

	Individual	Mean Vector
Group 1	1	(1.0, 1.0)
Group 2	4	(5.0, 7.0)

Step 2:

□ Thus, we obtain two clusters containing:

{1,2,3} and {4,5,6,7}.

□ Their new centroids are:

$$m_1 = \left(\frac{1}{3}(1.0 + 1.5 + 3.0), \frac{1}{3}(1.0 + 2.0 + 4.0) \right) = (1.83, 2.33)$$

$$m_2 = \left(\frac{1}{4}(5.0 + 3.5 + 4.5 + 3.5), \frac{1}{4}(7.0 + 5.0 + 5.0 + 4.5) \right)$$

$$= (4.12, 5.38)$$

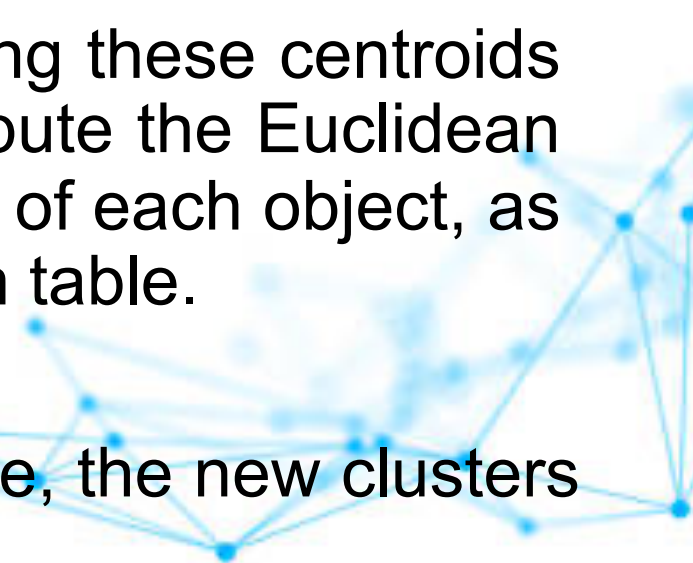
Individual	Centroid 1	Centroid 2
1		
2 (1.5, 2.0)		
3		
4		
5		
6		
7		

$$d(m_1, 2) = \sqrt{|1.0 - 1.5|^2 + |1.0 - 2.0|^2} = 1.12$$

$$d(m_2, 2) = \sqrt{|5.0 - 1.5|^2 + |7.0 - 2.0|^2} = 6.10$$

Step 3:

- Now using these centroids we compute the Euclidean distance of each object, as shown in table.
- Therefore, the new clusters are:
 $\{1,2\}$ and $\{3,4,5,6,7\}$
- Next centroids are:
 $m_1=(1.25,1.5)$ and $m_2 = (3.9,5.1)$



Individual	Centroid 1	Centroid 2
1		
2		
3		
4		
5		
6		
7		

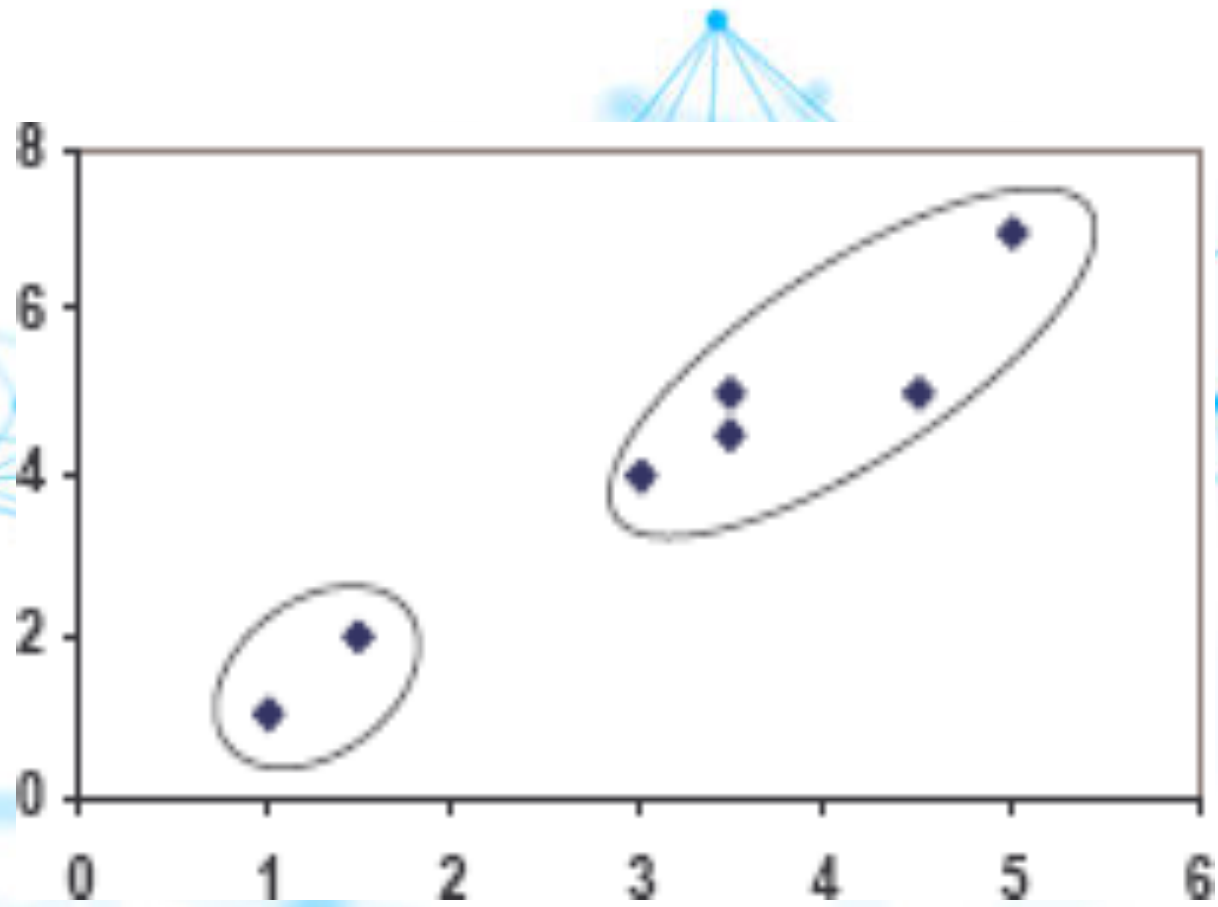
□ Step 4 :

The clusters obtained are:
 $\{1,2\}$ and $\{3,4,5,6,7\}$

- Therefore, there is no change in the cluster.
- Thus, the algorithm comes to a halt here and final result consist of 2 clusters $\{1,2\}$ and $\{3,4,5,6,7\}$.

Individual	Centroid 1	Centroid 2
1	0.58	5.02
2	0.58	3.92
3	3.05	1.42
4	6.68	2.20
5	4.18	0.41
6	4.78	0.61
7	3.75	0.72

PLOT



Example with K=3

Individual	$m_1 = 1$	$m_2 = 2$	$m_3 = 3$	cluster
1	0	1.11	3.81	1
2	1.12	0	2.5	2
3	3.81	2.5	0	3
4	7.21	8.10	3.81	3
5	4.72	3.81	1.12	3
6	5.31	4.24	1.80	3
7	4.30	3.20	0.71	3

C_3

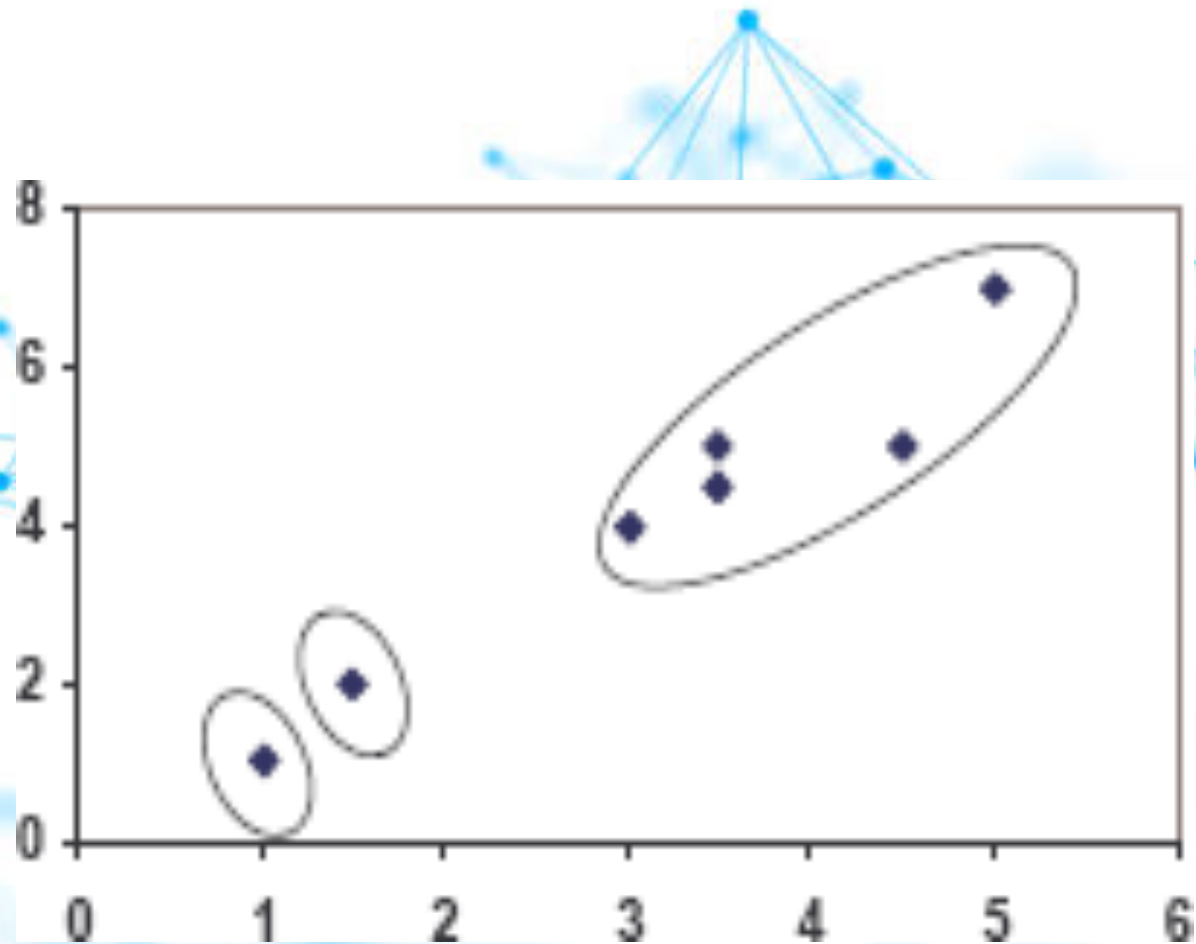
clustering with initial centroids (1, 2, 3)

Step 1

Individual	m_1 (1.0, 1.0)	m_2 (1.5, 2.0)	m_3 (3.9, 5.1)	cluster
1	0	1.11	5.02	1
2	1.12	0	3.92	2
3	3.81	2.5	1.42	3
4	7.21	8.10	2.20	3
5	4.72	3.81	0.41	3
6	5.31	4.24	0.81	3
7	4.30	3.20	0.72	3

Step 2

PLOT



How Many Clusters?

Number of clusters K is given

Partition n docs into predetermined number of clusters

Finding the “right” number of clusters is part of the problem

Given docs, partition into an “appropriate” number of subsets.

E.g., for query results - ideal value of K not known up

Tradeoff between having more clusters (better focus within each cluster) and having too many clusters

What is a Good Clustering?

Internal criterion: A good clustering will produce high quality clusters in which:

the intra-class (that is, intra-cluster) similarity is high
the inter-class similarity is low

The measured quality of a clustering depends on both the document representation and the similarity measure used

External criteria for clustering quality

Quality measured by its ability to discover some or all of the hidden patterns or latent classes in gold standard data

Assesses a clustering with respect to ground truth ... requires *labeled data*

Assume documents with C gold standard classes, while our clustering algorithms produce K clusters, $\omega_1, \omega_2, \dots, \omega_K$ with n_i members.

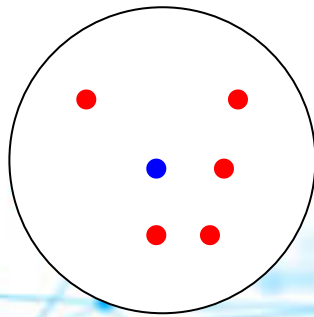
External Evaluation of Cluster Quality

Simple measure: purity, the ratio between the dominant class in the cluster ω_i and the size of cluster ω_i

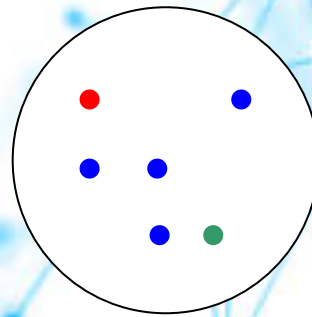
$$Purity(\omega_i) = \frac{1}{n_i} \max_j (n_{ij}) \quad j \in C$$

Biased because having n clusters maximizes purity
Others are entropy of classes in clusters (or mutual information between classes and clusters)

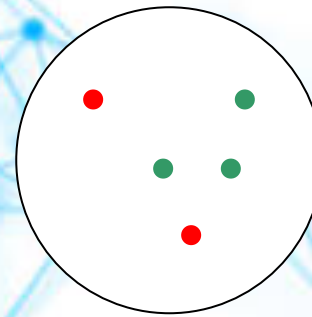
Purity example



Cluster
I



Cluster II



Cluster
III

Cluster I: Purity = $1/6 (\max(5, 1, 0)) = 5/6$

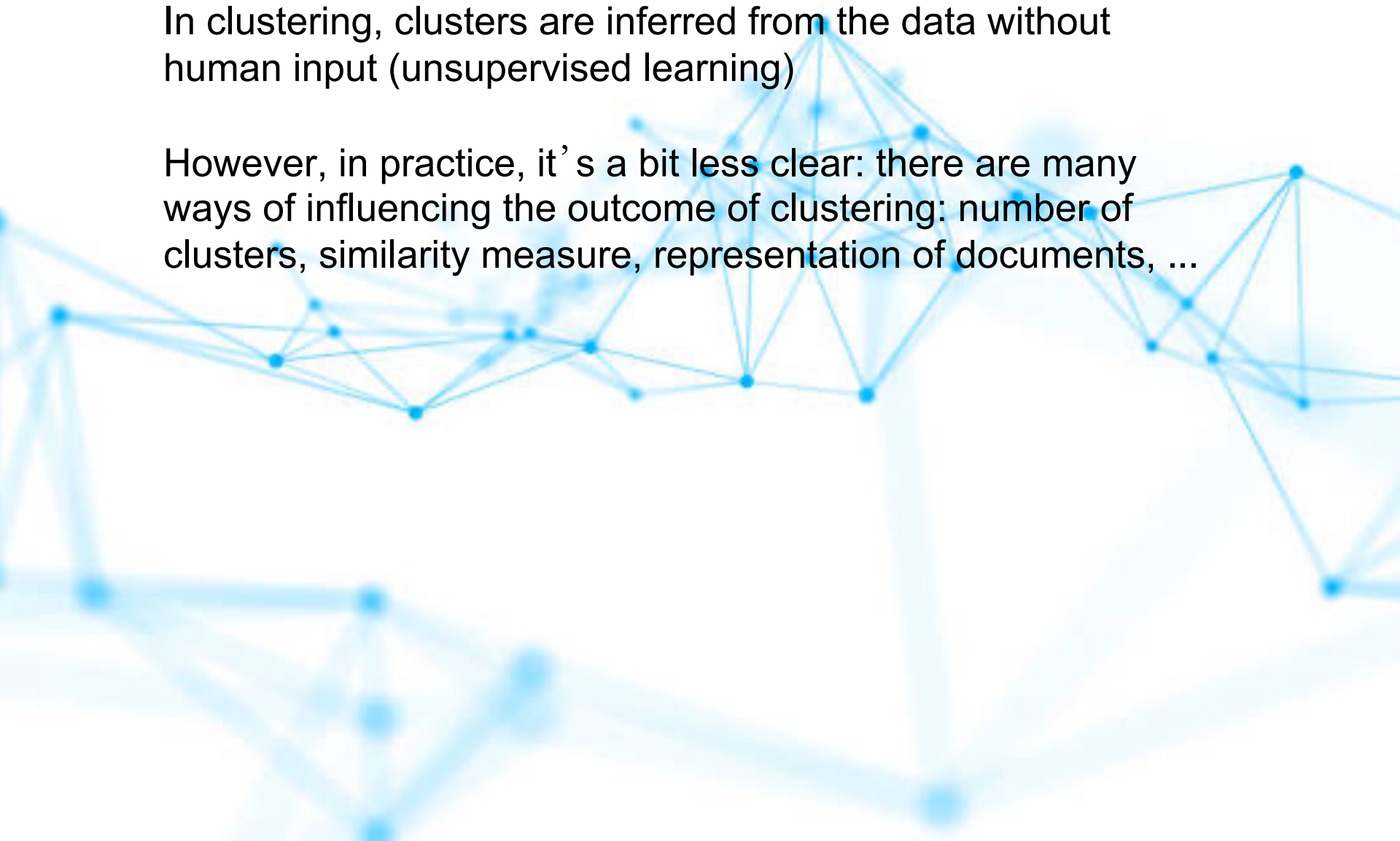
Cluster II: Purity = $1/6 (\max(1, 4, 1)) = 4/6$

Cluster III: Purity = $1/5 (\max(2, 0, 3)) = 3/5$

Final words

In clustering, clusters are inferred from the data without human input (unsupervised learning)

However, in practice, it's a bit less clear: there are many ways of influencing the outcome of clustering: number of clusters, similarity measure, representation of documents, ...



Assignment # 1

Assignment#1

Deadline is 19 FEB, 2023, 11:59 pm

